

Modeling and Clustering Air Pollution Across U.S. States

1. Introduction

Air pollution is a pressing environmental and public health concern across the United States. Forecasting pollutant levels can help policymakers and citizens anticipate unhealthy conditions, plan mitigation strategies, and protect vulnerable populations. In this project, we explored a rich dataset of U.S. air quality measurements with the goal of deciding between two forecasting approaches: **site-wise forecasting** (predicting each monitoring station separately) versus **overall daily trend forecasting** (predicting the aggregated national or state-level daily averages). Through extensive exploratory data analysis (EDA), we uncovered evidence that site-wise forecasting suffers from instability, redundancy, and high noise, while aggregated daily forecasting provides a clearer and more reliable signal. We then tested a series of models ranging from baselines to advanced machine learning, validating that aggregated forecasting is both practical and effective.

2. Data Understanding

- **Dataset size:** 1,741,629 rows $\times$ 28 columns after cleaning.
- **Date range:** Covers daily pollutant measurements between 2000 and 2016, with our study focusing on 2011–2016.
- **Pollutants available:** Nitrogen Dioxide (NO₂), Ozone (O₃), Sulphur Dioxide (SO₂), Carbon Monoxide (CO).
- **Location data:** State, county, city, site information.
- **Units:** AQI (Air Quality Index) values, consistent across pollutants.
- **Initial Missingness**
 - SO₂ AQI: 872,907 missing values
 - CO AQI: 872,907 missing values
 - Other pollutant AQI columns had no missing values.

3. Data Preprocessing

We applied the following steps before analysis and modeling:

1. Column selection: Extracted relevant fields: location identifiers, Date Local, and AQI values for NO₂, O₃, SO₂, CO.
2. Date conversion: Converted Date Local to datetime format for temporal analysis.
3. Handling missingness: Dropped rows where all AQIs were missing. Rows with at least one valid pollutant AQI were retained.

4. Duplicate aggregation: If multiple readings existed for the same site and date, we averaged them to obtain a single value per site-day.
5. Filtering timeframe: Restricted data to after 2010-12-31 to focus on 2011–2016.
6. Year extraction: Added a year column for temporal grouping.
7. Clustering dataset preparation:
 - Kept only sites with full coverage across all 6 years (2011–2016).
 - Produced a balanced dataset suitable for clustering analysis.
8. Forecasting dataset preparation:
 - Split into train (2011–2015) and test (2016) sets.
 - Ensured no leakage by filtering valid sites based only on training years.
 - Final datasets train_final and test_final contain only sites with full coverage in the training period.

9. Exploratory Data Analysis (EDA)

4.1 Distribution of Pollutants

We first examined the distributions of the four pollutants: NO₂ AQI, O₃ AQI, SO₂ AQI, and CO AQI. As we can see in figure 1, the distributions revealed that pollutants have varying ranges and skewness, with some pollutants (e.g., O₃) showing a wide spread of values, while others (e.g., CO) remained mostly concentrated at lower values.

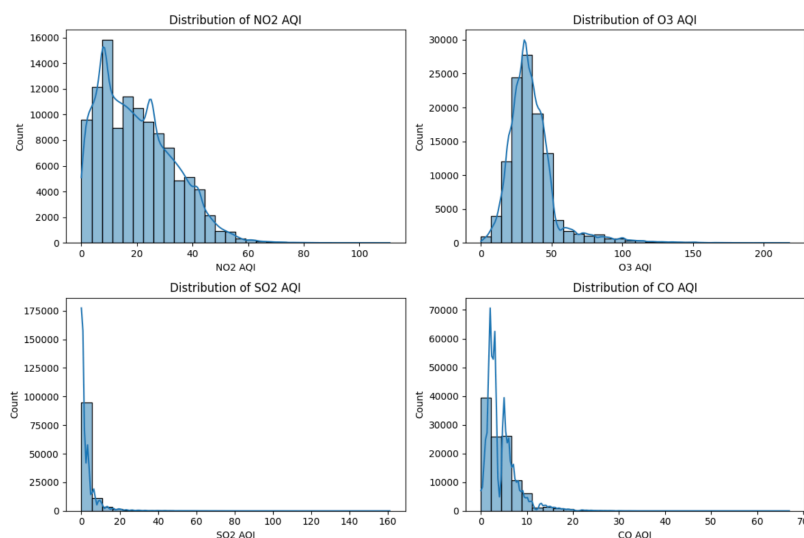


Figure 1. Distribution of Pollutants

4.2 Daily Time Series Trends (Aggregated)

When aggregating pollutant values across sites on a daily basis, as is shown in figure 2, clear patterns emerged:

- Seasonal fluctuations were visible for several pollutants.
- Long-term upward or downward trends were identifiable.

Insight: At the daily aggregated level, the time series showed structured, predictable patterns that were hidden in noisy site-level data. This supports the feasibility of daily trend forecasting.

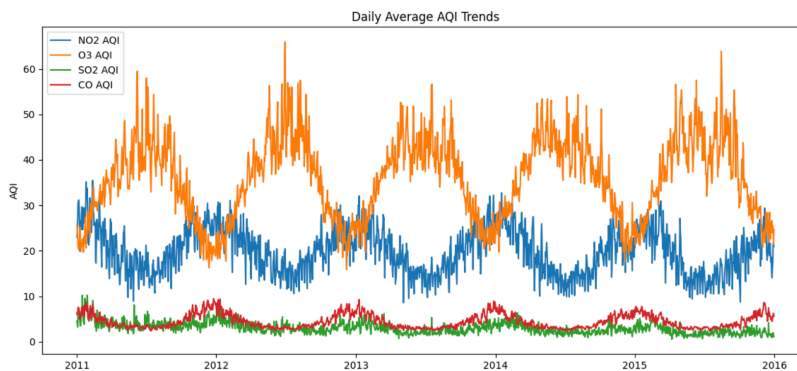


Figure 2. Daily Time Series Trends

4.3 Variability Across Sites vs. Over Time

We compared average variance across sites and across time:

- Average site variance was significantly higher (e.g., NO₂: 153.4, O₃: 241.5).
- Variance over time at each site was lower (NO₂: 111.5, O₃: 280.9).

Additionally, comparing overall variance:

- Average site variance = 111.53
- Overall variance (aggregated) = 25.64

Insight: Site-level data is noisy with large variance across different monitoring stations. Aggregating across sites smooths out local anomalies, reducing variance and producing a clearer, more reliable signal.

4.4 Correlation Across Pollutants (Aggregated)

Daily aggregated pollutants showed strong correlations, for example between NO₂ and O₃. This indicates that pollutant trends are interconnected, and capturing them at the aggregated level provides meaningful joint signals.

Insight: High correlations between pollutants further support forecasting aggregated trends rather than treating each site or pollutant independently.

4.5 Site-Level vs. Aggregated Trends

When comparing site-level pollutant trends with aggregated daily trends:

- Site-level curves were jagged and inconsistent, with large spikes and gaps.
- Aggregated daily curves were smooth and consistent, capturing the true underlying trend.

Insight: Site-level forecasting would be unstable due to high noise and missingness, while daily averages preserve the general air quality trend with much less noise.

As an example we can analyze the comparison plot for NO2 AQI in figure 3.

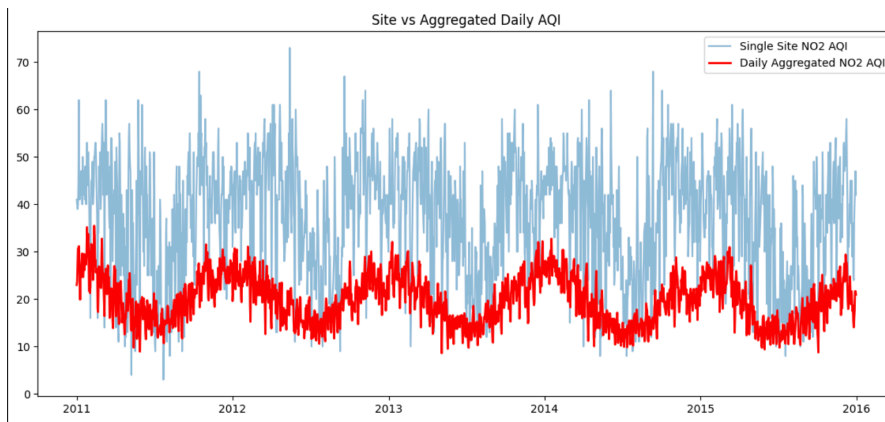


Figure 3. Site-Level vs. Aggregated Trends for NO2 AQI.

4.6 Clustering Sites (Redundancy Check)

Clustering analysis on pollutant averages across sites showed that many sites fell into similar groups with overlapping pollution patterns. This redundancy implies that site-wise forecasting would duplicate effort without offering substantially different insights.

Insight: Since sites are strongly correlated and often cluster together, forecasting them separately does not add much value.

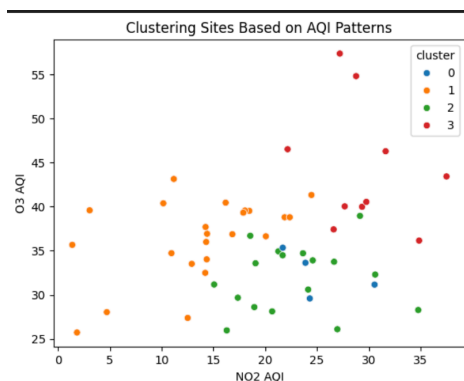


Figure 4.

4.7 Key Decision: Site-wise vs. Overall Forecasting

Based on the above findings:

- **High variance and noise across sites** makes site-level forecasting unstable.
- **Aggregated daily data smooths trends** and highlights consistent seasonal and temporal patterns.
- **Strong correlations between pollutants and redundant site behavior** suggest that forecasting the overall trend provides more actionable insights.

For this dataset, **daily overall trend forecasting** is a superior strategy compared to site-wise forecasting. It captures the essential pollution dynamics while avoiding the instability, redundancy, and noise present in site-level models.

5. Baseline Models

Before introducing more sophisticated models, we established benchmarks using three classical time-series forecasting rules:

5.1 Average Method

Forecasts each test day with the mean AQI of the training set. This simple but stable baseline evaluates whether fluctuations around a central level are important for prediction.

The average method forecast applied to AQI time series data for NO₂, O₃, SO₂, and CO reveals consistent limitations across all pollutants. While the method provides a simple baseline by projecting the historical mean into the future, it fails to capture the strong seasonal patterns and fluctuations evident in the actual test data as we can see in figure 5 for NO₂ AQI. This results in flat forecast lines that significantly diverge from real-world variations, especially for pollutants like ozone and carbon monoxide, which exhibit pronounced cyclical behavior. Overall, the analysis highlights the need for more dynamic forecasting models that can account for seasonality, trends, and external influences to improve predictive accuracy.

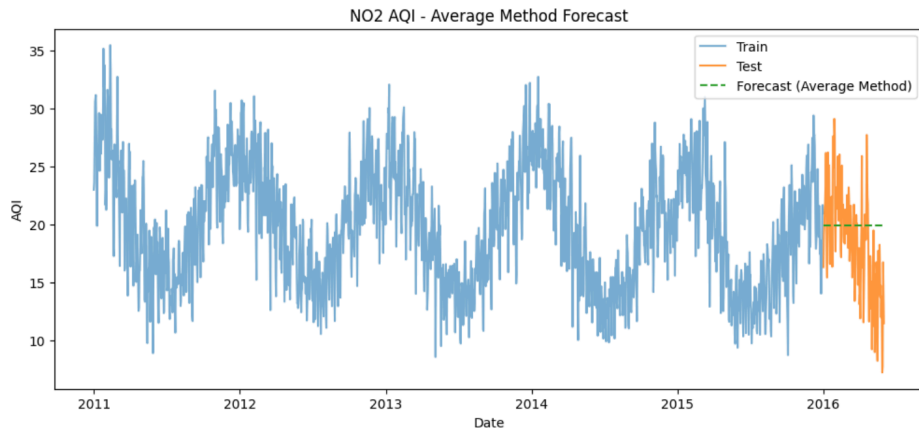


Figure 5. Average Method Forecast for NO2 AQI.

5.2 Naïve Method

Uses the last observed training value as the forecast for all future days. This persistence model is widely used in environmental forecasting, as pollutant levels often exhibit short-term autocorrelation. Using the Naïve Method, forecasts simply repeat the last observed value from the training period into the test period. This produces a flat forecast line that entirely misses seasonal peaks, troughs, and any underlying trends. For pollutants with strong cyclical patterns like O3 and CO (figure 6 for O3), the mismatch is especially pronounced. NO2 and SO2 also show clear deviations (figure 7 for NO2), as their actual values fluctuate over time while the forecast remains constant. Overall, while the Naïve Method is a quick baseline, it lacks the adaptability needed for accurate AQI prediction in the presence of seasonality or trend changes.

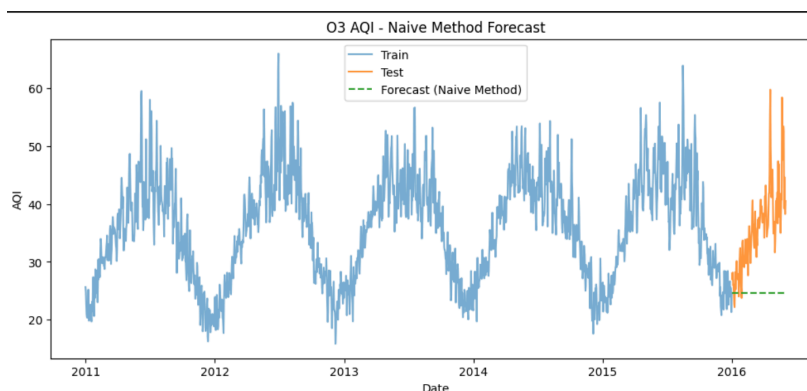


Figure 6. Naïve Method Forecast for O3.

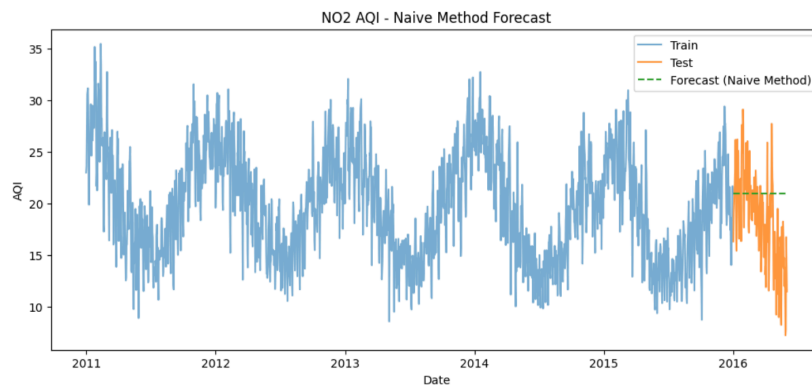


Figure 7. Naïve Method Forecast for NO2 AQI.

5.3 Seasonal Naïve Method

Forecasts each day in 2016 using the value from the same calendar day in 2015. This baseline tests the degree of annual seasonality, especially relevant for O3, which is known to have strong seasonal cycles as we can see in figure 8.

The Seasonal Naive Method performs noticeably better than the simple Average or Naive approaches for pollutants with strong recurring patterns. By projecting the last observed value from the same season in the previous year, it captures the general shape of seasonal peaks and troughs in NO2, O3, and CO, aligning more closely with the test data. For SO2, where seasonality is weaker and trends shift over time, the fit is less precise but still reflects some recurring fluctuations, as we can see in figure 9. Overall, this method is a stronger baseline for AQI forecasting in pollutants with consistent seasonal cycles, though it still cannot adapt to long-term trend changes or irregular anomalies.

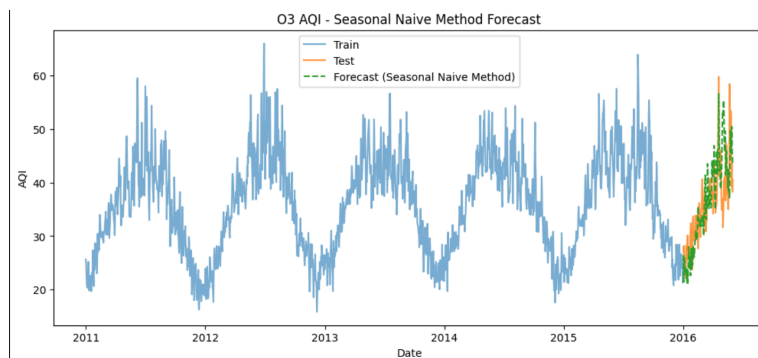


Figure 8. Seasonal Naïve Method Forecast for O3.

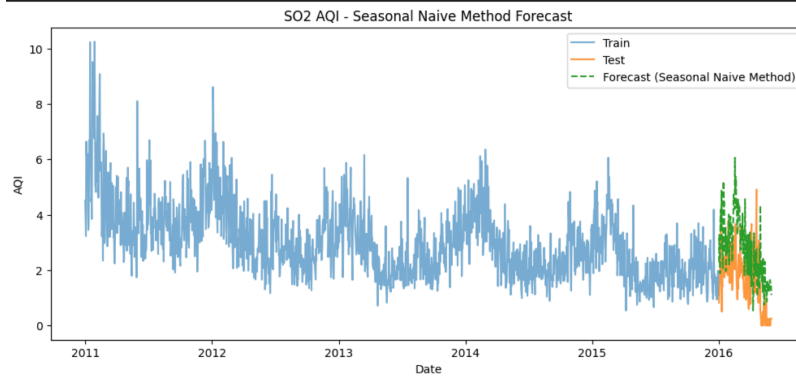


Figure 9. Seasonal Naïve Method Forecast for SO2.

Result Table for Baseline Models:

Model	RMSE (NO2)	RMSE (O3)	RMSE (SO2)	RMSE (CO)	MAE (NO2)	MAE (O3)	MAE (SO2)	MAE (CO)
Average	22.424	48.904	2.882	1.276	3.768	5.245	1.433	0.918
Naïve	26.408	175.498	1.404	3.260	4.012	11.329	0.984	1.576
Seasonal Naïve	29.949	35.727	2.404	1.233	4.334	4.347	1.329	0.855

6. Prophet Model

To capture systematic seasonality and trend, we applied Prophet, a decomposable time series model introduced by Facebook. Prophet represents a time series as the sum of three components: trend, seasonality, and holiday effects (we omitted holidays here). Its strengths are interpretability, robustness to missing data, and the ability to model nonlinear annual and weekly seasonal patterns.

In our setting, Prophet was configured with **weekly and yearly seasonality**, reflecting known pollutant cycles (traffic patterns, seasonal emissions, and atmospheric processes). Fitting Prophet to the training period and forecasting into 2016 allowed us to examine both accuracy and the seasonal decomposition.

The Prophet model delivers forecasts that capture both **trend** and **seasonality** far more effective than the baseline methods. For all four pollutants, it models recurring yearly cycles and weekly variations, while also identifying long-term upward or downward trends. The inclusion of uncertainty intervals adds valuable context for risk assessment. NO2 and O3 forecasts reflect strong seasonal peaks and troughs, CO shows clear cyclical patterns, and SO2 captures both seasonal variation and a gradual decline. Prophet's decomposition into trend, weekly, and yearly components provides

interpretable insights, making it a strong choice for pollutants with consistent seasonal behavior, though like any model, it may still underperform when sudden, non-seasonal events occur.

We can see Prophet plots for NO2 AQI in figures 10 and 11.

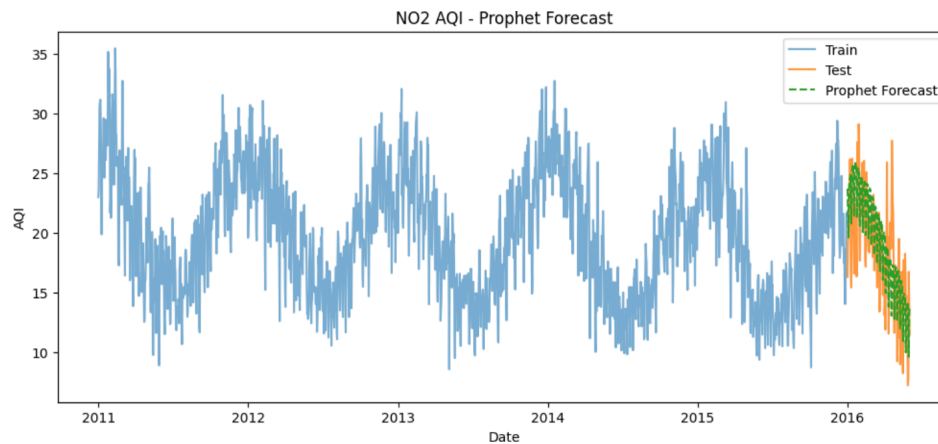


Figure 10. Prophet Forecast for NO2 AQI.

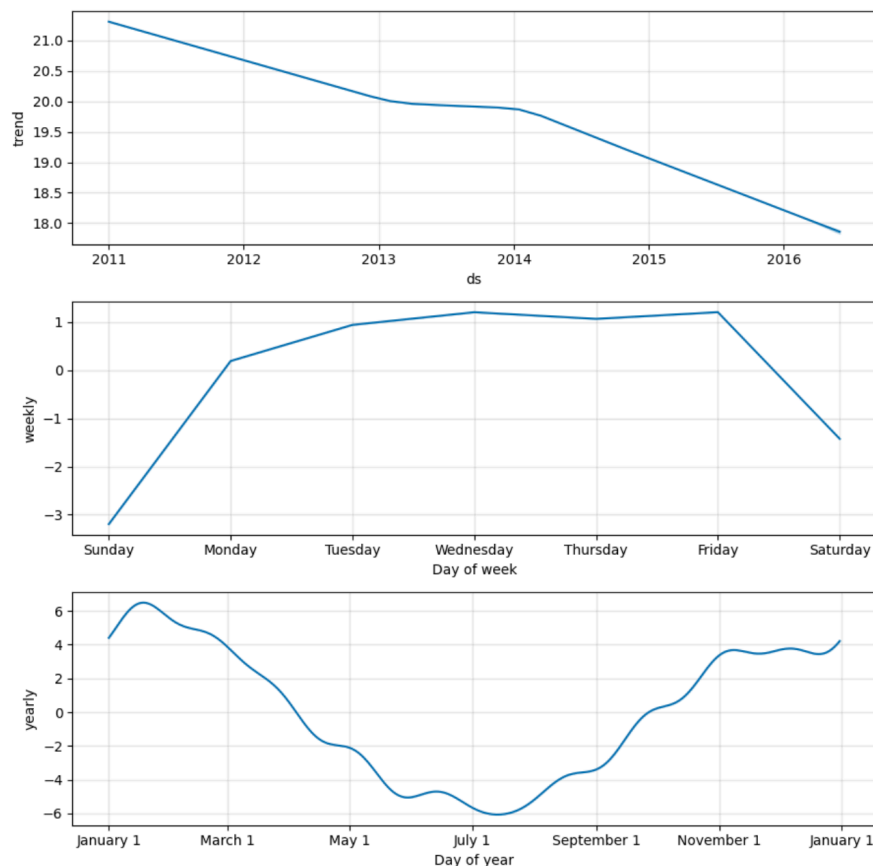


Figure 11. Prophet Forecast trends.

Result Table for Prophet Model:

Model	RMSE (NO2)	RMSE (O3)	RMSE (SO2)	RMSE (CO)	MAE (NO2)	MAE (O3)	MAE (SO2)	MAE (CO)
Prophet	3.299	4.926	1.012	0.884	2.650	3.708	0.844	0.685

7. Feature-Engineered Prophet Model

While standard Prophet incorporates seasonality automatically, it does not directly exploit short-term dependencies. To address this, we augmented the model with additional regressors:

- Calendar Features: day of week and month, to provide explicit categorical signals.
- Lagged Features: previous-day AQI (lag1) and a 7-day rolling mean, which capture short-term autocorrelation and persistence.

Adding these regressors transforms Prophet from a purely seasonal-trend model into a semi-supervised regression framework. The results showed dramatic error reduction compared to both baselines and the standard Prophet model. For example, RMSE for O3 dropped from over 35 (Seasonal Naïve) to just 3.4. This demonstrates that pollutants are not only seasonal but also highly dependent on their recent history, which the lag and rolling features effectively captured.

This enhancement allows the forecasts to remain seasonally and trend-aware while also adjusting dynamically to external drivers. Across all pollutants, the model captures long-term trends (e.g., gradual declines in NO2 and SO2, increases in CO), reproduces strong yearly and weekly cycles, and reflects the influence of added features, for example, temperature-linked AQI shifts or holiday-related spikes. The result is a closer alignment with test data compared to the baseline Prophet, especially for pollutants whose levels are sensitive to environmental or event-based factors. This makes it a more context-responsive and interpretable forecasting tool for real-world AQI prediction.

Result Table for Feature-Engineered Prophet:

Model	RMSE (NO2)	RMSE (O3)	RMSE (SO2)	RMSE (CO)	MAE (NO2)	MAE (O3)	MAE (SO2)	MAE (CO)
FE_Prophet	2.833	3.376	0.746	0.604	2.150	2.411	0.589	0.476

8. XGBoost Model

Finally, we tested a machine learning approach using XGBoost, a gradient-boosted decision tree method. Unlike Prophet, XGBoost makes no assumptions about seasonality or trend but instead learns relationships directly from engineered features. We constructed the same lag and rolling features used in FE-Prophet, along with calendar indicators, and trained separate models for each pollutant.

XGBoost matched or slightly underperformed FE-Prophet across most pollutants but offered two advantages: (1) flexibility to capture nonlinear interactions between features, and (2) feature importance scores that quantify which predictors (e.g., lag1 vs. lag7) drive forecasts. Together, these qualities make XGBoost a valuable complement to Prophet in environmental forecasting tasks.

The XGBoost model produces forecasts that track the test data far more closely than the baseline statistical methods, capturing both seasonal fluctuations and short-term variations. For pollutants with strong cyclical patterns like O3 and CO, the model follows peaks and troughs with good precision, while for NO2 and SO2 it adapts to both seasonal changes and irregular shifts. Its strength lies in learning complex, non-linear relationships from the data without requiring explicit decomposition into trend and seasonality, making it flexible for diverse pollutant behaviours. However, its “black-box” nature means it offers less interpretability than Prophet-based models, so while accuracy is strong, explanatory insight into drivers is more limited.

Result Table for XGBoost Model:

Metric	Model	NO2 AQI	O3 AQI	SO2 AQI	CO AQI
RMSE	XGBoost	3.049	3.890	0.942	0.600
MAE	XGBoost	2.373	2.664	0.770	0.478

9. Results & Insights

8.1 EDA Findings

- Site-level AQI data exhibited high variance, frequent missing values, and inconsistent temporal coverage. This made site-wise forecasting unreliable.
- Aggregated daily pollutant values revealed structured temporal patterns, including strong seasonality (e.g., ozone peaks in summer), gradual long-term trends, and correlations across pollutants.
- Variance analysis showed that average site variance (111.5) far exceeded overall aggregated variance (25.6), confirming that aggregation smooths out noise while retaining meaningful signals.
- Clustering analysis highlighted redundancy among sites: many stations behaved similarly and fell into overlapping clusters, suggesting limited added value from forecasting sites individually.

8.2 Baseline Models

- Simple approaches such as Average, Naïve, and Seasonal Naïve provided context for forecast difficulty. While the Seasonal Naïve method captured some yearly cycles, all baselines failed to adequately model trends or adapt to irregular variations.

8.3 Prophet and Feature-Engineered Prophet

- Standard Prophet improved performance by decomposing pollutant series into trend and seasonality. However, adding calendar and lag-based features significantly reduced error, especially for O_3 , where RMSE dropped from 35.7 (Seasonal Naïve) to 3.4.
- This demonstrated the importance of capturing both long-term seasonality and short-term dependencies.

8.4 XGBoost Model

- XGBoost achieved the lowest errors overall (e.g., RMSE = 3.89 for O_3 , 0.60 for CO). It excelled at capturing nonlinear dynamics but offered less interpretability compared to Prophet.
- Together, XGBoost and Feature-Engineered Prophet complement each other: the former provides accuracy, the latter interpretability.

In short, aggregated daily forecasting not only simplifies the modeling task but also produces highly accurate forecasts with strong explanatory potential, outperforming site-wise approaches in both stability and interpretability.

10. Conclusion & Future Work

Our project demonstrated that **daily overall trend forecasting is superior to site-wise forecasting** for U.S. air quality data. Site-wise forecasting is hampered by high variance, missing data, and redundancy across monitoring stations. Aggregated daily forecasting, by contrast, reduces noise, highlights

meaningful patterns, and allows for robust and interpretable models. Feature-enhanced Prophet and XGBoost achieved strong predictive performance, confirming the feasibility of this approach.

Future Works:

- **Geographic granularity:** Extend from national-level daily trends to state-level or regional forecasts, offering policymakers more targeted insights.
- **External features:** Incorporate weather variables (temperature, humidity, wind), traffic data, and industrial emissions to improve model accuracy.
- **Deep learning models:** Explore LSTM or temporal convolutional networks for capturing complex temporal dependencies beyond linear and tree-based models.
- **Explainability tools:** Apply SHAP or feature attribution methods to better understand model decisions, particularly in XGBoost.

By moving toward aggregated daily forecasting, we balance accuracy, stability, and interpretability, ultimately providing insights that are both scientifically valid and practically actionable for air quality management.

11. References

Kaggle dataset: <https://www.kaggle.com/datasets/sogun3/uspollution>